Homework 1

Module 1 - Introduction & Math Review · Homework · No date

Name _____ Date _____

1. A pharmacist dissolved 200 milligrams of a new antibiotic drug into a volume of purified water and placed the solution in a refrigerator (4°C). At various time intervals, the pharmacist removed a small aliquot from the solution and measured the amount of drug contained in each aliquot. The following data were obtained.

Time (hr)	Drug (mcg/mL)
0.5	233.3
1	217.6
2	189.5
4	143.6
8	82.5
12	47.4
16	27.2

1a Is the decomposition of this antibiotic a first-order or a zero-order process?

A. Neither: it needs a third plot
B. Zero order
C. First order

1b What is the rate constant k for the decomposition of this antibiotic?

1c What is the half-life $t_{1/2}$?

1d What was the initial volume of water used to prepare the original solution?

1e How much time is required for the concentration of drug to decrease by 75%?

2. Below is the decrease in the amount of drug as a function of time.

Time (hr)	Drug A (mg)
0.5	489
1	478
2	455
3	433
6	365
9	298
12	230
16	140

2a Does the decrease in the amount of drug A appear to be a zero-order or a first-order process?

A. First order
B. Zero order
C. Cannot tell without logarithms

2b What is the rate constant k?

2c What is the half-life t½?

3. A solution of a drug was freshly prepared at a concentration of 315 mg/mL. After 30

days at 25°C, the drug concentration in the solution was 75 mg/mL.

3a Assuming first-order kinetics, when will the drug decline to one-half of the original concentration?

3b Assuming zero-order kinetics, when will the drug decline to one-half of the original concentration?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

1a

Worked by the drill

Answer: C. First order

Over each 4-hour step from 4 to 16 hr the concentration falls by the same fraction (1.74, 1.74, 1.74), so ln C against time is a straight line. A constant fraction lost per equal time is first order, because the rate is proportional to the amount remaining.

Her homework with her own numbers; her key gives this answer.

1b

Worked by the drill

Answer: 0.1386 hr^{-1}

WHAT THE STEM GIVES

D_0	200 milligrams	*not needed: k comes from the slope, not the amount*
temperature	4°C	*not needed*
C_1	217.6 mcg/mL at 1 hr	$\ln(C_1)$, the first point
C_2	27.2 mcg/mL at 16 hr	$\ln(C_2)$, the second point
$t_2 - t_1$	16 hr − 1 hr	the denominator, 15 hr

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{\ln(C_1) - \ln(C_2)}{t_2 - t_1}, \text{ with } t_1 = 1 \text{ hr and } t_2 = 16 \text{ hr}$$

For first order, $\ln C = \ln C_0 - kt$, so ln C against t is a straight line with slope −k, where k is the first-order rate constant. Two points far apart give the slope with the least error; her key uses the 1 hr and 16 hr points, 15 hours apart.

ALGEBRA

$$k = \frac{\ln(217.6) - \ln(27.2)}{15 \text{ hr}} = \frac{2.0794}{15 \text{ hr}} = 0.1386 \text{ hr}^{-1}$$

$\ln(217.6) - \ln(27.2) = \ln(\frac{217.6}{27.2}) = 2.0794$: the mcg/mL cancel inside the logarithm, leaving a pure number. Dividing by the 15 hr between the points leaves reciprocal hours, so k = 0.1386 hr^{-1}, the fraction of the drug lost per hour.

> **DOES THE NUMBER MAKE SENSE?**
>
> From 1 hr (217.6) to 16 hr (27.2) the level falls eightfold, three halvings in 15 hours, so the half-life is near 5 hr and k near $\frac{0.693}{5}$; the two-point slope gives 0.1386 hr^{-1}.

Her homework with her own numbers; her key gives this answer.

1c Worked by the drill

Answer: 5 hr

WHAT THE STEM GIVES

D_0	200 milligrams	*not needed: a first-order half-life does not depend on the amount*
table	217.6 mcg/mL at 1 hr and 27.2 mcg/mL at 16 hr	gave k = 0.1386 hr^{-1} in part b, the denominator

THE WORKING, ONE LINE AT A TIME

SETUP

$$t½ = \frac{0.693}{k} = \frac{0.693}{0.1386\ hr^{-1}} = 5.00\ hr$$

The half-life t½ is the time for C to fall to half, so $e^{-k\,t½} = 0.5$ and $k\,t½ = \ln 2 = 0.693$, giving $t½ = \frac{0.693}{k}$. With k = 0.1386 hr^{-1} from part b, the reciprocal hours in the denominator invert to hours: 5.00 hr for every halving.

> **DOES THE NUMBER MAKE SENSE?**
> From 1 hr (217.6) to 16 hr (27.2) the level falls eightfold, three halvings in 15 hours, so the half-life is near 5 hr; $\frac{0.693}{0.1386} = 5.00$ hr agrees.

Her homework with her own numbers; her key gives this answer.

1d Worked by the drill

Answer: 800 mL

WHAT THE STEM GIVES

D_0	200 milligrams	written 200,000 mcg, the numerator
C_1	217.6 mcg/mL at 1 hr	brought back 1 hr to C_0 = 250.0 mcg/mL, the denominator

THE WORKING, ONE LINE AT A TIME

SETUP

$$\ln C_0 = \ln C + kt = \ln(217.6) + (0.1386)(1) \rightarrow C_0 = 250.0\ mcg/mL$$

The first-order line $\ln C = \ln C_0 - kt$ rearranges to $\ln C_0 = \ln C + kt$, with C_0 the concentration the moment the drug dissolved. The 1 hr point, 217.6 mcg/mL, has declined for 1 hour at k = 0.1386 hr^{-1}; adding kt = 0.1386 undoes that decline: C_0 = 250.0 mcg/mL.

UNIT

$$200\ mg \times 1000\ mcg/mg = 200{,}000\ mcg$$

The 200 mg dissolved is the amount at time zero. The concentration is in mcg/mL, so the amount is put in micrograms, 200 × 1000 mcg/mg = 200,000 mcg; then mcg over mcg/mL cancels the micrograms and leaves millilitres, the unit asked for the volume.

ALGEBRA

$$V = \frac{200{,}000 \text{ mcg}}{250.0 \text{ mcg/mL}} = 800 \text{ mL}$$

Concentration is amount over volume, $C_0 = \frac{D_0}{V}$, so $V = \frac{D_0}{C_0}$: the whole 200,000 mcg spread through the water at 250.0 mcg/mL. The micrograms cancel and millilitres are left, 800 mL, the volume of purified water the pharmacist used.

> **DOES THE NUMBER MAKE SENSE?**
> C_0 must be a little above the 0.5 hr value of 233.3 mcg/mL; it is 250.0. Then 200,000 mcg at 250.0 mcg/mL is $\frac{200{,}000}{250.0} = 800$ mL.

Her homework with her own numbers; her key gives this answer.

1e

Worked by the drill

Answer: 10 hr

WHAT THE STEM GIVES

D_0	200 milligrams	*not needed: the time to a fraction does not depend on the amount*
fraction	decrease by 75%	leaves 25%, so C_0 over C is 4, two half-lives
table	217.6 mcg/mL at 1 hr and 27.2 mcg/mL at 16 hr	gave $k = 0.1386$ hr^{-1} in part b

THE WORKING, ONE LINE AT A TIME

SETUP

$$t = \frac{\ln(C_0/C)}{k} = \frac{\ln(1/0.25)}{0.1386 \text{ hr}^{-1}}$$

The first-order line $\ln C = \ln C_0 - kt$ solved for the time is $t = \frac{\ln(C_0/C)}{k}$. A 75% decrease leaves 25% of the starting concentration, so C_0 over C = $\frac{1}{0.25} = 4$ whatever the starting value, and $k = 0.1386$ hr^{-1} is the rate constant from part b.

ALGEBRA

$$t = \frac{1.3863}{0.1386 \text{ hr}^{-1}} = 10.00 \text{ hr} = 2 \times 4.999 \text{ hr}$$

ln 4 = 1.3863, a pure number, so dividing by k in hr^{-1} leaves hours: $\frac{1.3863}{0.1386 \text{ hr}^{-1}} = 10.00$ hr. Falling to 25% is two halvings, 100% to 50% to 25%, so the time is exactly $2 \times t_{½} = 2 \times 4.999$ hr, which checks the answer against part c.

> **DOES THE NUMBER MAKE SENSE?**

> Losing 75% leaves a quarter, two halvings, so the time is two half-lives: 2 × 4.999 hr = 10.00 hr, more than the one half-life of 4.999 hr.

Her homework with her own numbers; her key gives this answer.

2a Worked by the drill

Answer: B. Zero order

Each 3-hour step from 3 to 12 hr loses about the same 67–68 mg, so the amount against time is a straight line on ordinary axes. A constant amount lost per equal time is zero order, because the rate does not depend on how much drug is left.

Her homework with her own numbers; her key gives this answer.

2b Worked by the drill

Answer: 22.5 mg/hr

WHAT THE STEM GIVES

A1	478 mg at 1 hr	the first amount
A2	230 mg at 12 hr	the second amount
$t_2 - t_1$	12 hr − 1 hr	the denominator, 11 hr

THE WORKING, ONE LINE AT A TIME

SETUP

$$k_0 = \frac{A1 - A2}{t_2 - t_1}$$

For zero order the amount falls by the same quantity each hour, $A = A0 - k_0\, t$, so k_0, the zero-order rate constant, is the constant amount lost per unit time, the slope of amount against time: the difference between two amounts over the time between them, in mg/hr.

ALGEBRA

$$k_0 = \frac{478\ \text{mg} - 230\ \text{mg}}{12\ \text{hr} - 1\ \text{hr}} = \frac{248\ \text{mg}}{11\ \text{hr}} = 22.55\ \text{mg/hr}$$

478 mg at 1 hr minus 230 mg at 12 hr is 248 mg lost over 12 − 1 = 11 hr, the two points her key uses; $\frac{248\ \text{mg}}{11\ \text{hr}} = 22.55$ mg/hr. Milligrams over hours is an amount per unit time, the unit of a zero-order rate constant.

ROUND

$$k_0 = 22.5\ \text{mg/hr}$$

Her key gives 22.5 mg/hr, the 22.545 mg/hr rounded to one decimal place. The table is not perfectly straight: the 1 hr and 16 hr points give 22.53 mg/hr, the same to one decimal place.

> **DOES THE NUMBER MAKE SENSE?**
> From 3 hr (433 mg) to 6 hr (365 mg) the drop is about the same as from 9 to 12 hr, 298 to 230 mg, a nearly constant loss; 478 − 230 = 248 mg over 11 hr is 22.5 mg/hr.

Her homework with her own numbers; her key gives 22.5 mg/hr, the 22.545 mg/hr rounded to one decimal place.

2c Worked by the drill

Answer: 11.1 hr

WHAT THE STEM GIVES

A at 1 hr	478 mg	brought back 1 hr to A0 = 500.5 mg, rounded to 500 mg
table	478 mg at 1 hr and 230 mg at 12 hr	gave k_0 = 22.5 mg/hr in part b, in the denominator $2k_0$

THE WORKING, ONE LINE AT A TIME

SETUP

$$A0 = A + k_0\,t = 478\text{ mg} + (22.545\text{ mg/hr})(1\text{ hr}) = 500.5\text{ mg}$$

The zero-order line $A = A0 - k_0\,t$ rearranges to $A0 = A + k_0\,t$. Her key starts from the 1 hr entry, 478 mg, which has already lost (22.545 mg/hr)(1 hr) = 22.545 mg; adding it back gives A0 = 500.5 mg, which her key rounds to 500 mg.

ALGEBRA

$$t½ = \frac{A0}{2k_0} = \frac{500\text{ mg}}{2(22.5\text{ mg/hr})} = 11.11\text{ hr}$$

Half of A0 is 250 mg, and at 22.5 mg/hr that loss takes $\frac{250\text{ mg}}{22.5\text{ mg/hr}} = 11.11$ hr, which is $t½ = \frac{A0}{2k_0}$. Unlike first order, this half-life grows with the starting amount, because a constant loss takes longer to remove half of a larger amount.

ROUND

$$t½ = 11.1\text{ hr}$$

Her key gives 11.1 hours. The unrounded 500.5 mg and 22.545 mg/hr give $\frac{500.5\text{ mg}}{2(22.545\text{ mg/hr})} = 11.10$ hr, the same to one decimal place.

> **DOES THE NUMBER MAKE SENSE?**
> Half of A0 = 500 mg lies between the 298 mg at 9 hr and the 230 mg at 12 hr, so t½ is between 9 and 12 hr: 11.1 hr.

Her homework with her own numbers; her key gives 11.1 hours, from A0 rounded to 500 mg and k_0 to 22.5 mg/hr.

3a Worked by the drill

Answer: 14.5 days

WHAT THE STEM GIVES

C_0	315 mg/mL	the numerator of the ratio inside the logarithm
t	30 days	the denominator of k
temperature	25°C	*not needed*
C	75 mg/mL	the denominator of the ratio inside the logarithm

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{\ln(315/75)}{30\ \text{days}} = \frac{1.4351}{30\ \text{days}} = 0.04784\ \text{day}^{-1}$$

First order: $\ln C = \ln C_0 - kt$, so $k = \frac{\ln(C_0/C)}{t}$, the log of the ratio of starting to final concentration over the time between them. $\frac{315}{75} = 4.200$, a pure number because the mg/mL cancel, and $\ln 4.200 = 1.4351$; the time is in days, so k is in day^{-1}.

ALGEBRA

$$t½ = \frac{0.693}{0.04784\ \text{day}^{-1}} = 14.49\ \text{days}$$

$t½ = \frac{0.693}{k}$ because $k\ t½ = \ln 2$ when C falls to half; the reciprocal days in k invert to days: $\frac{0.693}{0.04784\ \text{day}^{-1}} = 14.49$ days. A first-order half-life does not depend on the starting concentration: every halving takes the same 14.49 days.

ROUND

$t½ = 14.5$ days

Her key gives 14.5 days, the 14.49 days rounded to one decimal place. Under first order every halving takes this long, whatever the concentration it starts from.

DOES THE NUMBER MAKE SENSE?

315 to 75 mg/mL is a fall to under a quarter in 30 days, more than two halvings, so t½ is under 15 days: 14.5 days.

Her homework with her own numbers; her key gives this answer.

3b Worked by the drill

Answer: 19.7 days

WHAT THE STEM GIVES

C_0	315 mg/mL	in k_0 and the numerator of $\frac{C_0}{2k_0}$
t	30 days	the denominator of k_0
temperature	25°C	*not needed*
C	75 mg/mL	subtracted from C_0 in k_0

THE WORKING, ONE LINE AT A TIME

SETUP

$$k_0 = \frac{315 - 75\ \text{mg/mL}}{30\ \text{days}} = 8.0000\ \text{(mg/mL)/day}$$

Zero order: $C = C_0 - k_0 t$, so $k_0 = \frac{C_0 - C}{t}$, the constant loss per unit time. $315 - 75 = 240$ mg/mL was lost over 30 days, so $k_0 = 8.0000$ (mg/mL)/day, a concentration per unit time rather than the day^{-1} of a first-order k.

ALGEBRA

$$t_{½} = \frac{C_0}{2k_0} = \frac{315\ \text{mg/mL}}{2(8.0000\ (\text{mg/mL})/\text{day})} = 19.69\ \text{days}$$

Half of the original 315 mg/mL is 157.5 mg/mL; at a constant 8.0000 (mg/mL)/day that loss takes $\frac{157.5\ \text{mg/mL}}{8.0000\ (\text{mg/mL})/\text{day}} = 19.69$ days, which is $t_{½} = \frac{C_0}{2k_0}$. The mg/mL cancel and days are left; the zero-order half-life uses the starting concentration, so it differs from the first-order 14.49 days of part a.

ROUND

$t_{½} = 19.7$ days

Her key gives 19.7 days, the 19.69 days rounded to one decimal place. The same two points give 14.5 days under first order, so the order decides the half-life.

DOES THE NUMBER MAKE SENSE?

Zero order loses the same amount each day. In 30 days more than three-quarters of the 315 mg/mL was lost, so halving takes less than two-thirds of 30 days, under 20 days: 19.7 days.

Her homework with her own numbers; her key gives this answer.